Technical White Paper

# The Federated Harbor:

## Identity, Coordination, and Settlement Across Administrative Domains

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**Abstract**

A demonstration is scheduled for 4pm. Alice runs the back end on her laptop; Bob runs the front end on his. Each operator has a working local harbor in the sense of the two prior papers in this series — capability tokens are pinned and attenuated, evidence is Merkle-chained, bonds are posted, and the Conservation invariant holds inside each machine. The problem is not internal. Alice's `db:write` card is gibberish to Bob's daemon, Bob's revocation gossip is invisible to Alice's mesh, and a bond Alice posted to cover a botched migration sits in her harbor's collateral pool while the migration lands in Bob's staging database. Two harbors, each individually trustworthy, jointly useless. This paper extends the Anchor Protocol's capability ceremony, the Bonded Commons' evidence trail and competitive-insurance market, and the daemon-as-correlation-device argument to settings where the sovereign is not single. We define an inter-harbor capability transfer protocol with explicit attenuation across the boundary; a federated revocation gossip layer with a named convergence bound and cross-witness equivocation detection; a settlement protocol in which a bond posted at one harbor clears against damage measured at another through a structurally bounded escrow; and an admission ceremony for new harbors to join an existing mesh without prior reputation. We are disciplined about which claims are mechanized, which are bounded against named threat windows, and which are honestly open. The point of the paper is to draw the new boundary cleanly so that the unsolved problems are visible rather than hidden.

**Keywords:** federated identity, capability delegation, cross-realm authorization, mechanism design, distributed trust, certificate transparency, atomic settlement, mutual suspicion.

*Reading time: approximately 45 minutes end-to-end. The companion papers,* The Anchor Protocol *(Owens 2026) and* The Bonded Commons *(Owens 2026), supply the cryptographic and economic primitives this paper extends. Either of those can be read first or in parallel; this paper is the federation half. The Reader's Map below provides shorter routes for specific audiences.*

## Reader's Map

The paper has a topology: it opens with the gestalt (§1–§2), then moves through three load-bearing primitives in order of cryptographic depth (capability transfer, revocation, settlement), then steps back to admission, failure modes, and limitations. The figures are designed to be readable in sequence without the surrounding prose; if you want the five-figure summary, that path is given below.

| Layer | Load-bearing artifact | Lives at |
|---|---|---|
| **(1) Topology** | Two harbors, witness log, settlement escrow as a four-element gestalt | §2; Figure 1 |
| **(2) Cap. transfer** | Four-message ceremony; ProVerif model of cross-realm authenticity & attenuation, *conditional on a witness-log freshness assumption* (§4.5) | §4; Figure 3 |
| **(3) Revocation** | Federated cuckoo-filter mesh; $\Delta(1 + \ln m)$ convergence bound | §5; Figure 4 |
| **(4) Settlement** | Bounded escrow with two-outcome decision space; cross-harbor Conservation | §6; Figure 5 |
| **(5) Threat bands** | What is mechanized, what is bounded, what is open | §12; Figure 2 |

**Suggested reading paths by reader type:**

- **Beginner / new to the series.** Read §1 (the two-machine problem) and §8 (Alice and Bob, end-to-end). The worked example is written to be read cold. Look at the five figures in order. Skip the proofs on the first pass.
- **Distributed-systems engineer.** §2 (the topology), §4 (transfer ceremony), §5 (gossip and equivocation). The companion Anchor paper for the cryptographic substrate. Then §12 for the honest open questions.
- **Formal-methods reviewer.** Jump to §9 (the TLA+ and ProVerif extensions) and Appendix A (the verification status table). Threat bands in Figure 2 are the most concentrated signal of what is and is not mechanized.
- **Cryptoeconomic / mechanism-design reader.** §6 (bounded escrow) and §7 (cold-start admission via bonded sponsorship). The proposal's equilibrium contributions are stated here as open work; that is honest, not coy.
- **Security-engineering reviewer.** §3 (threat model) first, then the three primitive sections in order. §10 for the operationally interesting cases the protocols do not close.

**Companion papers.** The Anchor Protocol pins down what it means for a process on a single machine to prove its identity to the local daemon. The Bonded Commons explains why coordination at scale needs a pre-transactional commons authority rather than per-transaction trust negotiation. This paper extends both into the cross-machine, cross-organization case. We will cite specific sections (e.g. *Anchor §4*, *Bonded §??*) when we depend on a primitive rather than re-derive it.

# Contents

# 1 Introduction: The Two-Machine Problem

A demonstration is scheduled for 4pm. Alice runs the back end of the demo on her laptop; Bob runs the front end on his. Each has a working Port Daddy daemon: capability tokens are minted and attenuated under the Anchor Protocol [1], the workspace history is Merkle-chained and replicated locally per the Bonded Commons [2], and bonds for cleanup of botched migrations are posted in the local collateral pool. Inside each laptop, the trust story is closed and the formal-verification artifacts hold.

The demo nevertheless does not work. Three failures, none of them inside either machine:

1. Alice's agent obtains a `db:write` card for the staging database, attempts to use it against Bob's daemon, and is refused. Bob's daemon has never seen Alice's signing key. The card is well-formed under the Anchor Protocol, but Bob has no root of authority for it; from his daemon's point of view, the card is gibberish.

2. Five minutes before the demo, Alice's operator revokes the `db:write` card after spotting that it has been over-attenuated. The revocation propagates within Alice's mesh in under a second by the gossip protocol of Anchor §4. Bob's harbor never hears about it. The card remains live on Bob's side.

3. The migration runs anyway, lands in Bob's staging database, and corrupts the demo schema. Alice posted a bond in her harbor against exactly this case. The bond sits in her harbor's collateral pool. Bob's harbor measured the damage. Neither harbor can, on its own, route the bond to where the damage is.

Each failure is a different facet of the same underlying gap. Both daemons are internally consistent. Neither is a root of authority for the other. The single-machine sovereign of the prior papers does not extend to the case of two machines under two operators with no shared administrative parent.

We call this the **two-machine problem**. The point of stating it this baldly is that the failure modes are not exotic; they are what every developer who has tried to dovetail two local agent fleets on a shared task has seen. The bug is structural. The bug is not in either daemon. The bug is in the assumption that one daemon's authority extends past the machine boundary by default. It does not, and it should not.

> *The trick is not to extend a single sovereign across machines. The trick is to admit there are two and to specify what they owe each other.*

This paper extends the Anchor Protocol, the Bonded Commons, and the daemon-as-correlation-device argument to the multi-administrative-domain case. We are not introducing a new substrate; we are showing that the existing substrate admits a clean federation, and that the federation rules are sharper than they look. Where the prior papers spoke of *the* commons authority, this paper speaks of a *mesh of* mutually witnessing commons authorities, each sovereign inside its own machine, none sovereign over the others, and all of them bound to the same audit substrate.

## 1.1 What this paper is and is not

This paper is the third in a tightly-coupled series. The Anchor Protocol is the cryptographic-token substrate; the Bonded Commons is the economic and governance argument; this paper is the federation layer. The relationships are concrete: §4 extends Anchor's Phase-3 delegation chain across a trust boundary; §5 extends Anchor's cuckoo-filter mesh; §6 extends Bonded's Conservation invariant.

What this paper is *not*: it is not a permissionless-blockchain redesign. Port Daddy daemons do not run consensus. The Federated Harbor explicitly rejects the assumption common to most modern federation proposals [24, 23] that a shared ledger is the right substrate. The cost of that simplification is that we cannot lean on a chain for ordering, equivocation detection, or settlement; we must construct each of those primitives from peer gossip plus a federated witness log. The advantage is that the design composes cleanly with the local-first, single-operator-machine architecture of the prior two papers.

### 1.2 Contributions

The contributions are concrete. We separate them by mechanization status to be honest about which are theorems, which are bounded threat models, and which are sketches.

1. **The four-element federation topology (§2).** A precise statement of the boundary between intra-harbor and inter-harbor scope. Defines harbor sovereignty as a property characterization in the style of Bonded's Admissible KMS, so that subsequent claims can be checked against it.

2. **The inter-harbor capability transfer ceremony (§4).** A four-message protocol exchanging a Harbor Card issued by $A$ for an attenuated card valid at $B$, without either daemon trusting the other's signing key as a root. The ceremony binds the issuing harbor's current epoch root into the envelope, which is the substrate that revocation in §5 hooks into. ProVerif model in Appendix B (status: **partial**, model drafted; soundness pending mechanization).

3. **Federated revocation mesh (§5).** The multi-administrative-domain extension of Anchor's cuckoo-filter gossip protocol. Each harbor publishes an epoch root to a shared witness log; cross-witness signatures generalize the Certificate Transparency auditor pattern [9] to capability tokens. We prove a $\Delta(1 + \ln m)$ convergence bound on revocation propagation for $m$ federated harbors gossiping every $\Delta$, and an $O(\log m)$ expected detection bound on cross-witness equivocation (Theorem 5.1).

4. **Cross-harbor settlement (§6).** A protocol for atomic clearing of a bond posted at $A$ against damage measured at $B$, with a third-party escrow whose role is structurally bounded: it can refuse to clear, but it cannot redirect funds, equivocate about clearing decisions, or extract more than a pre-agreed fee. The bounded-escrow design is inspired by HTLC-style atomic-swap constructions [15] but specialized to a setting where the assets are reputation-priced under competitive insurance, not fungible coins.

5. **Cold-start admission ceremony (§7).** A protocol for a new harbor joining an existing mesh without prior reputation, using a bonded-sponsor pattern: an existing harbor posts a bond on the newcomer's behalf, which is forfeit on misbehavior in a probationary epoch. This is the federation-layer analogue of competitive insurance from Bonded.

6. **Honest open problems (§12).** Three named open questions: (a) whether the correlated-equilibrium reframing of Bonded survives the multi-principal setting cleanly; (b) whether bonds can settle without a trusted-but-bounded escrow third party; (c) whether the folk-theorem cartel-resistance argument from Bonded weakens at the federation layer, and by how much. We state these as open, not as solved.

The proposal that scoped this paper [3] listed two further contributions (a formal definition of harbor sovereignty, and equilibrium results across federations co-authored with Youle) which are present here in skeletal form and will be load-bearing in a subsequent revision once the open problems above resolve. We will not claim a result we do not have.

## 2 The Federated Authority

The Bonded Commons paper defined a *commons authority* as a pre-transactional trust infrastructure with sovereign scope inside a single machine. Inside that scope, the daemon is the unique correlation device that turns mutually-suspicious agents into a correlated equilibrium [2]. The single-machine sovereign was an honest simplification. We now drop it.

The federation we build has four elements (Figure 1). Two harbors, each sovereign inside its own machine. A shared witness log to which each harbor publishes its current epoch root, and which any third party can audit for equivocation. A settlement escrow with a structurally bounded decision space. Every primitive in the rest of the paper attaches to one of these four elements.

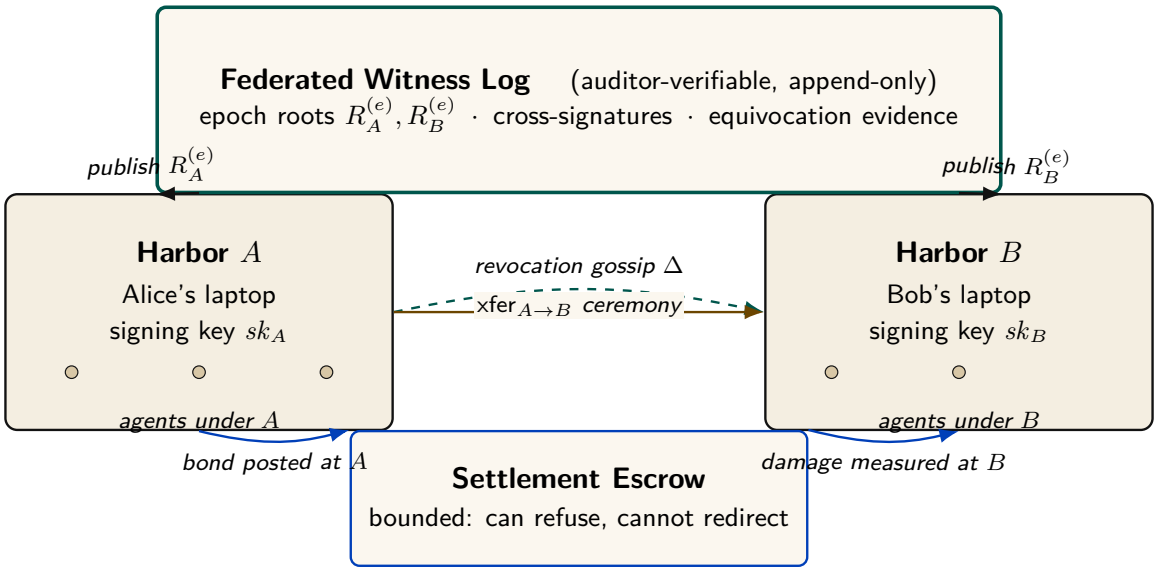


Figure 1: The Federated Harbor as a four-element gestalt. Neither harbor is a root of authority for the other. The witness log above is the shared correlation device — each harbor publishes its epoch root, and any third party can audit for equivocation. The escrow below is structurally bounded (it can refuse to clear, but it cannot redirect funds or equivocate about its decisions). The amber arrow between the harbors is the cross-harbor capability transfer ceremony of §4; the dashed sage line is the federated revocation gossip of §5. The diagram is deliberately small: every primitive in the paper attaches to one of these four elements.

### 2.1 Harbor sovereignty as a property characterization

In Bonded, the Admissible KMS was characterized by a list of properties rather than by a specific construction (per-machine signing key, per-machine SQLite, etc.) so that the argument carried over to any concrete instantiation. We adopt the same posture here.

**Definition 2.1** (Harbor Sovereignty)**.** A daemon $D_A$ is *sovereign over harbor* $A$ if it has exclusive authority over five things:

1. **Issuance.** Only $D_A$ can mint a Harbor Card whose root signature verifies under $A$'s signing key $sk_A$.
2. **Attenuation.** Only $D_A$ can apply a local attenuation predicate $\text{att}_A$ that restricts an inbound capability before it is verified by an $A$-side agent.
3. **Revocation.** Only $D_A$ can decide that a card it issued is revoked. The decision becomes globally visible by the gossip protocol of §5; $D_A$ does not control when other harbors learn of it, but it controls whether the revocation event exists.
4. **Acceptance.** $D_A$ alone decides which inbound cards from other harbors to accept and under what local policy. No external party can compel acceptance.

5. **Evidence binding.** The epoch root $R_A^{(e)}$ that $D_A$ publishes to the witness log is signed under $sk_A$ and binds the entire $A$-side state of issued, attenuated, and revoked cards up to epoch $e$. No other harbor can publish a root that purports to be $A$'s.

This is a deliberately narrow definition. It does *not* say that $D_A$ is sovereign over everything an $A$-side agent does; the agent may be operating against a database hosted at $B$, in which case $B$-side policy applies and $D_A$ has nothing to say about it. Sovereignty is over the *issuance and revocation surface of A's own cards*, not over downstream effects.

We will use Definition 2.1 as the discipline against which subsequent protocols are checked. The transfer ceremony of §4 produces a card valid at $B$; if it required $B$ to verify under $sk_A$ in the hot path, it would violate $B$'s sovereignty by making $B$'s acceptance contingent on an external authority. We will see that it does not.

## 2.2 Inter-harbor scope

The dual notion is also load-bearing. *Inter-harbor scope* is the scope of operations whose effects cross the trust boundary between sovereign harbors.

**Definition 2.2** (Inter-Harbor Scope)**.** An operation $o$ has *inter-harbor scope* $(A, B)$ if its execution causes a state change observable to $B$ under authority that originated at $A$. Examples: an $A$-issued card used to write to a $B$-hosted database; a settlement clearing a bond posted at $A$ against damage measured at $B$; a revocation issued by $A$ becoming visible to a $B$-side verifier.

Most operations are *intra-harbor*: minting, verifying, attenuating, and revoking cards under one harbor's authority, observed by agents under that same harbor. The Anchor Protocol's proofs cover the intra-harbor case end-to-end. The Federated Harbor's contribution is exactly the inter-harbor surface: every load-bearing protocol in this paper is one that lives at the boundary.

## 2.3 The witness log

The witness log is the device that makes the federation auditable without making any harbor sovereign over the others. It plays the same role that the Merkle Forest plays inside a single machine in Bonded (§6 of the companion paper): a tamper-evident accumulator that any third party can verify without needing to trust the publisher.

**Property 2.1** (Witness-Log Admissibility)**.** A witness log $W$ is admissible for the Federated Harbor if it provides:

1. **Append-only.** Once $W$ records an entry, $W$ cannot retract or reorder it.
2. **Per-harbor latest-root binding.** For each harbor $X$ in the federation and each epoch $e$, $W$ records at most one root $R_X^{(e)}$ signed under $sk_X$.
3. **Auditable consistency.** Any third party can verify, given $R_X^{(e_1)}$ and $R_X^{(e_2)}$ with $e_1 \le e_2$, that $R_X^{(e_2)}$ is a Merkle-consistent extension of $R_X^{(e_1)}$.
4. **Equivocation detectability.** If a malicious daemon $D_X$ presents two different roots $R_X^{(e)}$ and $R_X^{(e)\prime}$ to two different parties, any auditor that observes both detects the equivocation in expected $O(\log m)$ gossip rounds where $m$ is the size of the federation (Theorem 5.1).
5. **Threshold publishability.** The act of publishing $R_X^{(e)}$ to $W$ does not require $D_X$ to trust any other party; the cross-signatures collected from other daemons in the federation are an auditor convenience, not an authorization gate.

This list is intentionally close in shape to the five properties of the Admissible KMS in Bonded (the "Federated Sovereign" section of the companion paper). The Bonded paper used the same posture — characterizing the property surface, then exhibiting a concrete construction that

satisfied it — and we follow that pattern. Section 9 sketches one concrete instantiation built on Certificate Transparency-style append-only logs [9] cross-witnessed in the Sigstore/Rekor pattern [10]; the substrate need not be unique.

### 2.4 What the federation is not

It is worth saying what the federation is not, so that the rest of the paper does not have to keep refuting positions it never took.

The federation is *not* a consensus protocol. There is no global agreement on the order of events; there is per-harbor monotonicity (epoch roots only extend), there is cross-harbor witnessing (each root is published to a shared log), and there is detectable equivocation. None of these require harbors to agree on a global order, and the prior papers' designs explicitly avoid that overhead.

The federation is *not* a hub-and-spoke topology with a privileged coordinator. The witness log is replicable; in practice a small set of mutually-witnessing log instances suffices, and any harbor can read from any of them. The hub-and-spoke failure mode — one well-funded coordinator captures the federation by becoming everyone's only witness — is a real risk we discuss in §10, and the structural pushback is gossip-based witness redundancy.

The federation is *not* a permissionless market in identities. New harbors enter through the admission ceremony of §7, which requires either a pre-existing bond or an existing harbor sponsor. This is by design; the cost is that the federation is invitation-bounded, the benefit is that Sybil attacks against the federation [16] cost the attacker as much per identity as honest entry does.

## 3 Threat Model

We make the threat model explicit before introducing the protocols so that the design choices can be checked against named adversary powers. We follow the Dolev–Yao convention [7] extended to the federation case: an attacker controls every wire between harbors, can intercept and replay any cross-machine message, and may collude with any bounded number of harbor operators. The attacker cannot break the underlying cryptographic primitives (Ed25519 signatures, SHA-256 hashes); these assumptions are inherited from Anchor.

### 3.1 In-scope attacks

The protocols in this paper are designed against the following attackers.

**Cross-realm identity spoofing.** An attacker presents a card at $B$ claiming issuance by $A$, using a forged signature under $sk_A$. *Closed by* signature verification against $A$'s public key as recorded in the witness log (§4). Reduces to Ed25519 EUF-CMA security [8].

**Replay across the boundary.** An attacker captures a valid `xfer_offer` from $A$ to $B$ at epoch $e$ and replays it at epoch $e+k$. *Closed by* the binding of $R_A^{(e)}$ into the envelope; the verifier at $B$ checks that $R_A$ is the current root, and a stale envelope is rejected (§4). The window between issuance and the next epoch is the structural attacker window, named and bounded.

**Capability inflation across transfer.** An attacker presents at $B$ a card $C'_B$ that claims authority beyond the attenuation predicate applied at $A$. *Closed by* the requirement that the verifier at $B$ recompute the composed attenuation $\mathrm{att}_B \circ \mathrm{att}_A$ from the envelope rather than trust an intermediate claim (§4); reduces to Anchor's intra-harbor attenuation invariant.

**Stale-revocation acceptance.** An attacker at $B$ presents a card that has been revoked at $A$ but not yet propagated to $B$. *Bounded by* the convergence theorem of §5: the attack window is at most $\Delta(1 + \ln m)$ rounds, and the bond posted at $A$ is sized to cover damage within that window.

**Cross-witness equivocation.** A malicious daemon $D_A$ publishes one root $R_A^{(e)}$ to one witness instance and a contradictory root $R_A^{(e)\prime}$ to another. *Bounded by* the auditor pattern of §2.3; expected detection in $O(\log m)$ rounds, with the witness bond at $A$ slashed on confirmation.

**Settlement redirection or equivocation.** The escrow attempts to redirect funds, equivocate about its clearing decision, or extract more than the pre-agreed fee $\phi$. *Closed structurally* by Theorem 6.1: the protocol restricts the escrow's decision space to two outcomes (clear or refuse) and prices its maximum extractable fee.

**Harbor membership forgery at the federation boundary.** An attacker presents a card from a harbor purporting to be in the federation but that has no admission record. *Closed by* the admission ceremony (§7) which produces a permanent, witness-logged enrollment record co-signed by an existing harbor's sponsor bond. A claimed harbor with no enrollment record is treated as outside the federation regardless of how well-formed its cards appear.

### 3.2 Out-of-scope attacks

We are explicit about what is not closed.

**Compromise of a harbor's signing key.** If $sk_A$ leaks, the attacker can issue arbitrary $A$-side cards until $A$ rotates the key and publishes the rotation to the witness log. We inherit Anchor's key-rotation procedure unchanged. The federation does not close the underlying primitive; it does bound the blast radius to $A$'s sovereign surface, which is exactly the right scope.

**Sybil at the federation layer.** An attacker who can pay the admission cost $m$ times can run $m$ federated harbors. The cost ratio between honest and adversarial admission is the load-bearing parameter; we discuss it in §7 but do not give a tight bound. This is named open work.

**Cartel formation across harbors.** Bonded's folk-theorem cartel-resistance argument relied on the daemon as a correlation device inside one machine. Across harbors, cartels can form offline and present a unified front to the witness log; the witness log records that all harbors agreed, which is consistent with both honest agreement and cartel collusion. We acknowledge this and state the open work in §12.

**Centralization gravity.** Historically, federated identity systems concentrate on one hub within five to ten years [17]. The Federated Harbor's structural pushback — replicable witness logs, no central coordinator — is design discipline, not theorem. We name this honestly in §10.

**Side channels.** Constant-time signature verification, cache-side-channel resistance, and timing-attack resistance are inherited from the underlying libraries; we do not re-prove them and we note where we trust them.

### 3.3 Threat-band visualization

Figure 2 lays out the threat surface as three bands: mechanized claims (teal), bounded threats (amber), and acknowledged open frontier (cobalt). The boundary between amber and cobalt shifts over time as future work mechanizes what is presently bounded. We will refer back to this figure when discussing limitations.

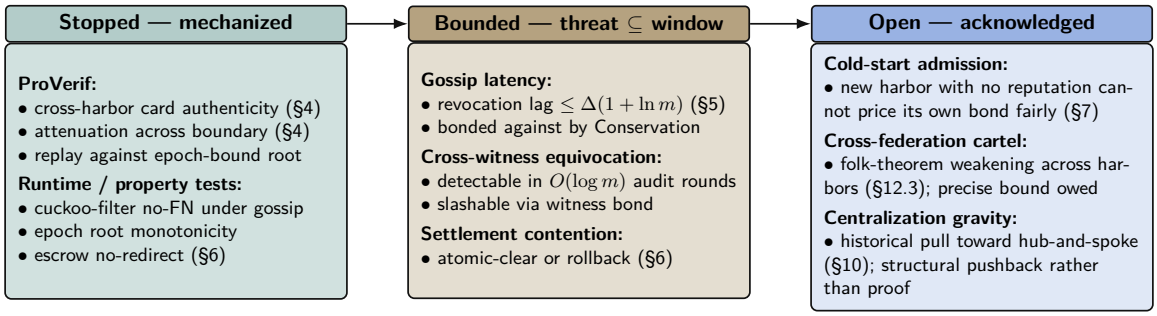


Figure 2: Threat-band layout for the Federated Harbor. The Anchor paper's defense-in-depth structure carries over: every claim either has a mechanization artifact (teal), a named bound that the bond can price against (amber), or an explicit acknowledgement of the open frontier (cobalt). The paper is disciplined about which band each claim lives in — see Appendix A for the artifact registry. Reading left to right runs deeper into threat space and out of formal reach; bands shift left over time as future work mechanizes what is presently bounded.

## 4 Cross-Harbor Capability Transfer

The first load-bearing protocol is the cross-harbor capability transfer ceremony. An $A$-side agent holds a Harbor Card $C_A$ minted by $D_A$ under Anchor Phase 2 or Phase 3. It needs to present an equivalent capability at $B$. The Anchor Protocol's intra-harbor delegation already supports attenuation; we extend that to the case where the attenuation crosses the trust boundary.

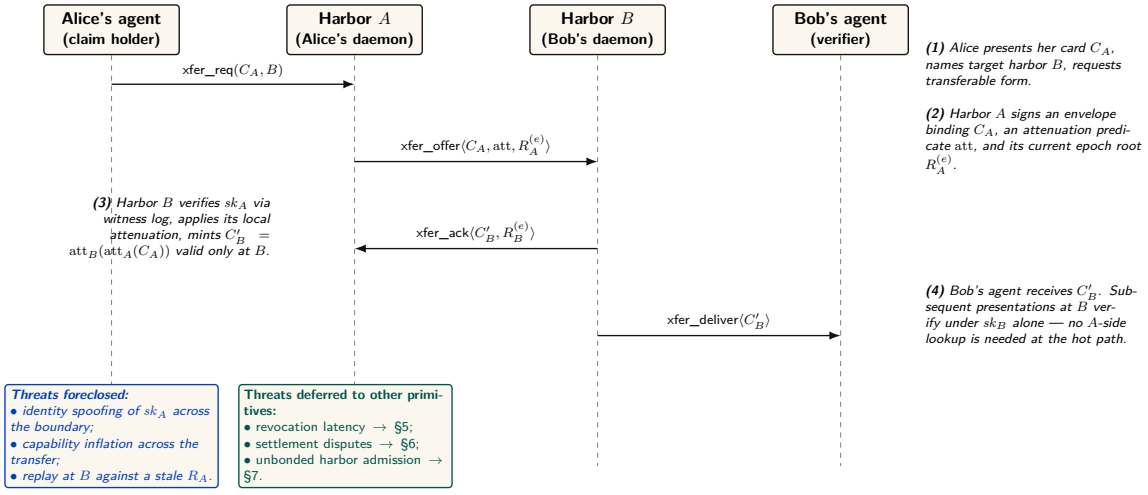


Figure 3: The four-message cross-harbor capability transfer ceremony. The critical property is that no message after (3) needs to reach back to Harbor $A$ in the hot path: $C'_B$ is verifiable under $sk_B$ alone, which is why federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the load-bearing primitive — a revocation issued by $A$ after epoch $e$ invalidates $C'_B$ as soon as the new $R_A^{(e+1)}$ reaches the witness log that $B$ audits.

### 4.1 The four-message ceremony

Figure 3 shows the protocol as a sequence diagram. The four messages are:

1. xfer_req$(C_A, B)$ — Alice's agent presents $C_A$ to its local daemon $D_A$ and names $B$ as the target harbor. The request is intra-harbor; standard Anchor verification applies.
2. xfer_offer$\langle C_A, \mathrm{att}_A, R_A^{(e)} \rangle$ — $D_A$ constructs an envelope binding the card, an attenuation predicate $\mathrm{att}_A$ specifying what subset of $C_A$'s authority is being offered for cross-realm use, and its current epoch root $R_A^{(e)}$. The envelope is signed under $sk_A$ and sent to $D_B$.
3. xfer_ack$\langle C'_B, R_B^{(e)} \rangle$ — $D_B$ verifies the envelope's signature against $A$'s public key as recorded in the witness log; verifies that $R_A^{(e)}$ is the current root for $A$; applies its own local attenuation $\mathrm{att}_B$; and mints $C'_B = \mathrm{att}_B(\mathrm{att}_A(C_A))$, a card valid only at $B$, signed under $sk_B$. The acknowledgment is sent back to $D_A$ along with $B$'s current epoch root.
4. xfer_deliver$\langle C'_B \rangle$ — $D_B$ delivers $C'_B$ to the agent at $B$ that will use it. Subsequent presentations of $C'_B$ at $B$ verify under $sk_B$ alone — no $A$-side lookup is needed at the hot path.

The critical property is that after message (3), no further synchronous interaction with $A$ is required for the agent to operate at $B$. This is what keeps the federation from collapsing into RPC-per-authorization. The cost is that revocation of $C_A$ at $A$ does not instantaneously revoke $C'_B$ at $B$; the propagation is gossip-bounded by Theorem 5.1 and the structural attacker window is named in the threat model.

### 4.2 Why this composes cleanly with Anchor

The Anchor Protocol's Phase-3 delegation already proves that a chain $C_{\text{daemon}} \to C_A \to C_B$ of attenuated cards is verifiable under the issuing daemon's public key, and that an attacker cannot inflate a downstream cap to exceed an upstream one. The transfer ceremony is Phase-3 delegation across a trust boundary, with two changes:

1. The second link in the chain is signed under a *different* root key ($sk_B$ instead of $sk_A$), so the verifier at $B$ verifies under $sk_B$, not $sk_A$.
2. The cross-realm link binds the issuing harbor's current epoch root $R_A^{(e)}$, so a revocation at $A$ after epoch $e$ invalidates the downstream card as soon as the new $R_A^{(e+1)}$ is observed at the verifier.

The composition is what makes the ceremony cheap to add: we are not introducing a new cryptographic primitive, only a new policy at the boundary that says "the verifier accepts a card whose chain crosses the trust boundary if and only if (a) each cross-realm link is signed under the receiving harbor's key and (b) the issuing harbor's bound epoch root is the current one in the witness log."

### 4.3 Attenuation across the boundary

Two attenuation predicates are applied in sequence. $\mathrm{att}_A$ is set by $D_A$ at issue time and reflects what authority $A$ is willing to project across the boundary. $\mathrm{att}_B$ is set by $D_B$ at acceptance time and reflects what authority $B$ is willing to grant on its own resources to a card whose root is $A$. Both attenuations strictly narrow the underlying capability.

**Lemma 4.1** (Attenuation Monotonicity Across the Boundary)**.** For any cross-realm card $C'_B = \mathrm{att}_B(\mathrm{att}_A(C_A))$, the set of operations authorized by $C'_B$ is a subset of those authorized by $C_A$ under the federated authorization rule.

*Sketch.* Each attenuation is a subset operation in the Anchor algebra; subset composition is a subset. Cross-realm policy at $B$ adds the constraint "and the operation must be permitted by $B$'s local policy," which can only further restrict. See the Anchor companion paper's capability-attenuation section for the intra-harbor argument; the cross-realm case adds no new mathematics, only a new applicable attenuation step. □

### 4.4 Threats foreclosed and threats deferred

The annotation band at the bottom of Figure 3 summarizes which threats this ceremony forecloses and which are deferred to other primitives. Specifically:

- **Foreclosed:** identity spoofing of $sk_A$ across the boundary; capability inflation across the transfer; replay at $B$ against a stale $R_A$.
- **Deferred:** revocation latency (handled in §5); settlement disputes (handled in §6); admission of unbonded new harbors (handled in §7).

We are explicit that the ceremony does not close every threat in one step. Federation is a layered defense; this is one layer.

### 4.5 Mechanization status

We have a draft ProVerif model of the four-message ceremony (Appendix B), extending the Anchor Phase-3 model with a second signing key and an epoch-root binding. The current status of the cross-realm authenticity query is **partial**: the model checks, but the proof depends on an assumption about witness-log freshness that we are still tightening. We mark this honestly in Appendix A and treat the attenuation monotonicity claim (Lemma 4.1) as proved under the same caveat.

## 5 Federated Revocation

The second load-bearing protocol is the federated revocation mesh. The Anchor Protocol's cuckoo-filter gossip handles revocation inside a single mesh by anti-entropy reconciliation [14]: each daemon maintains a local cuckoo filter $F$ of revoked card IDs, and pairs of daemons periodically reconcile. The Federated Harbor extends this to the multi-administrative-domain case.

Two pieces are added on top of the Anchor mechanism: (a) per-epoch roots $R_X^{(e)}$ published to a shared witness log so a third party can audit any harbor's local view, and (b) cross-witness signatures so the auditor pattern generalizes Certificate Transparency [9].

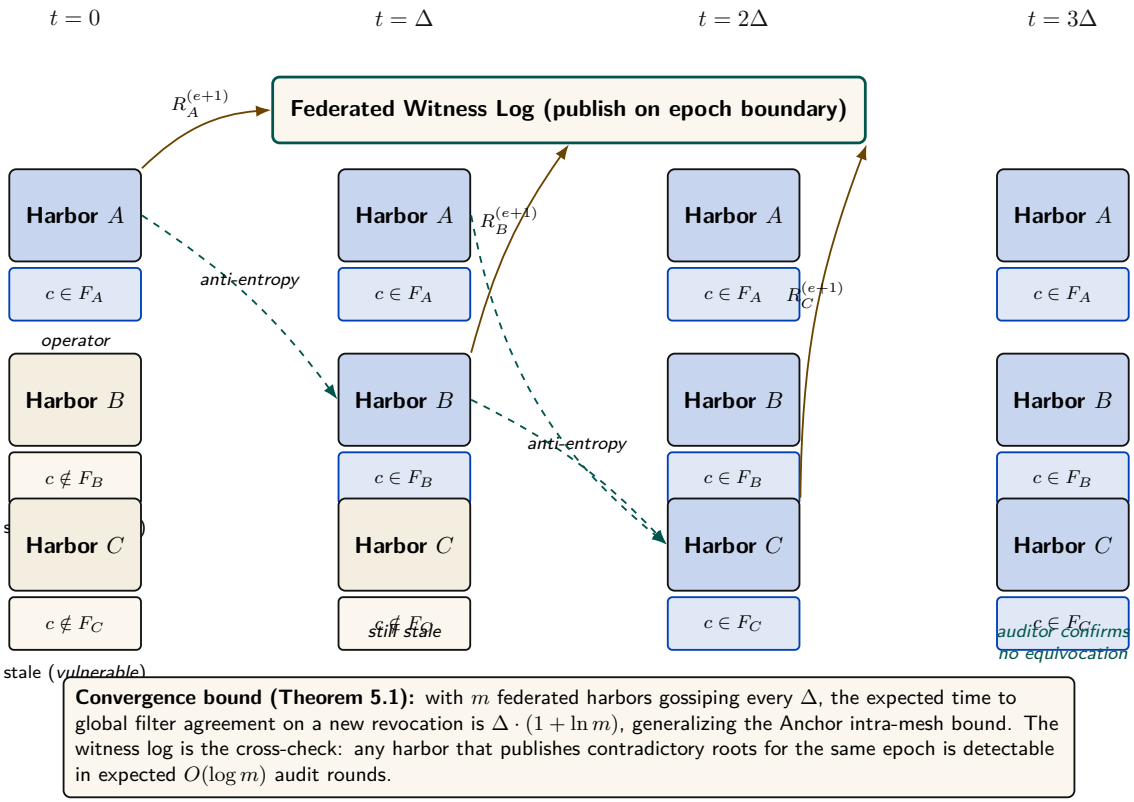


Figure 4: Federated revocation propagation across three harbors. The cuckoo filter mechanism is unchanged from Anchor §?? of the companion paper, but two new pieces are bolted on: (a) per-epoch roots $R_X^{(e)}$ published to a shared witness log so a third party can audit any harbor's local view, and (b) cross-witness signatures so the auditor pattern generalizes Certificate Transparency [9]. The window between $t = 0$ and $t = \Delta$ is the structural attacker window: bounded, named, and priced into the bond in §6.

## 5.1 Convergence bound

Anchor's intra-mesh convergence bound for cuckoo-filter anti-entropy is well known [14]: with $m$ daemons gossiping every $\Delta$ time units, the expected time for a new revocation to reach all daemons is $O(\Delta \log m)$. The Federated Harbor inherits this bound at the federation level, with the modification that each harbor's filter is treated as one node in the gossip graph (regardless of how many local daemons participate in the harbor's mesh).

**Theorem 5.1** (Federated Revocation Convergence)**.** Let $m$ be the number of federated harbors, each running an Anchor cuckoo-filter mesh with gossip period $\Delta$. Suppose harbor $X$ revokes card $c$ at time $t_0$ and publishes the resulting $R_X^{(e+1)}$ to the witness log at time $t_0 + \delta_W$ where $\delta_W \leq \Delta$. Then:

1. **Filter convergence.** The expected time for $c$ to appear in every federated harbor's local filter is $\Delta(1 + \ln m)$.
2. **Witness-log convergence.** Any auditor monitoring the witness log observes $R_X^{(e+1)}$ within $\delta_W$.
3. **Cross-witness equivocation detection.** If $D_X$ attempts to publish contradictory roots $R_X^{(e+1)}$ and $R_X^{(e+1)\prime}$ to distinct auditors, the contradiction is detected in expected $O(\log m)$ audit rounds.

*Sketch.* Part (1) is the standard anti-entropy bound from Demers et al. [14] applied at the harbor granularity. Part (2) is immediate from the publish step. Part (3) follows from Certificate

Transparency's standard argument [9]: $\log m$ auditors gossiping their observed roots detect a contradictory pair with probability approaching 1 in expected $O(\log m)$ rounds. We do not claim novelty for the underlying mathematics, only for the federation-layer assembly. □

**Corollary 5.1** (Bounded Attacker Window)**.** An attacker who obtains a card revoked at $X$ at time $t_0$ has a structurally bounded window of $\Delta(1 + \ln m)$ in which to present the card at a harbor that has not yet observed the revocation. The bond posted at $X$ for the card is sized to cover damage within this window (§6).

The bound is named and load-bearing. It is the device that makes the cross-harbor settlement protocol financially closed: the bond is priced to absorb damage during the structurally bounded propagation window, not to prevent damage from occurring at all. This is the same defense-in-depth posture as Bonded's bond-prices-damage argument [2], lifted to the federation layer.

### 5.2 Conflict resolution across harbors

The interesting case is disagreement. Harbor $A$ has revoked card $c$; harbor $B$ has not yet seen the revocation; an agent presents $c$ at $B$ for an action that affects $A$'s database. Three candidate resolution rules from the proposal [3], evaluated:

1. **Pessimistic (anyone-revokes-is-revoked).** The card is revoked iff any harbor has revoked it. Defended against by an attacker who would gain by injecting spurious revocations into the gossip mesh; rejects too many honest cards in equilibrium.
2. **Authoritative (only-issuer-revokes).** Only $A$ can revoke $A$'s cards. This is the rule we adopt. Combined with the convergence bound, it gives a clean attacker window and a clean settlement story.
3. **Epoch-bounded (revocation propagates within $\Delta$ epochs and inconsistency is a contract violation).** A refinement of (2) for the case where one harbor falls persistently behind. We treat this as a degraded mode of (2) rather than a separate rule.

We adopt rule (2). The pessimistic rule has the wrong defaults; the epoch-bounded rule is operationally the same as (2) with a watchdog. Adopting (2) means the Conservation invariant of §6 can be stated cleanly.

### 5.3 Cuckoo filter discipline

The cuckoo filter retains all Anchor-side disciplines (false-positive rate within $2B/2^f$ at load 0.95, pollution resistance under gossip) and adds one: a filter at $B$ that holds $c \in F_B$ where $c$ was revoked at $A$ is correct only if there is a corresponding inclusion proof in $R_A^{(e+1)}$ in the witness log. False positives that arise from gossip noise are detected and corrected; spurious revocations injected by an attacker into a peer's filter are rejected on first attempt to verify against the witness log. The discipline is: *local filters are advisory; the witness log is authoritative.*

## 6 Cross-Harbor Settlement

The third load-bearing protocol is cross-harbor settlement. A bond is posted at $A$ to cover damage caused by an $A$-side agent operating at $B$. When damage occurs, the bond must clear to the damaged party at $B$. The protocol must do this without making either harbor sovereign over the other, and without admitting a trusted third party in the operational sense.

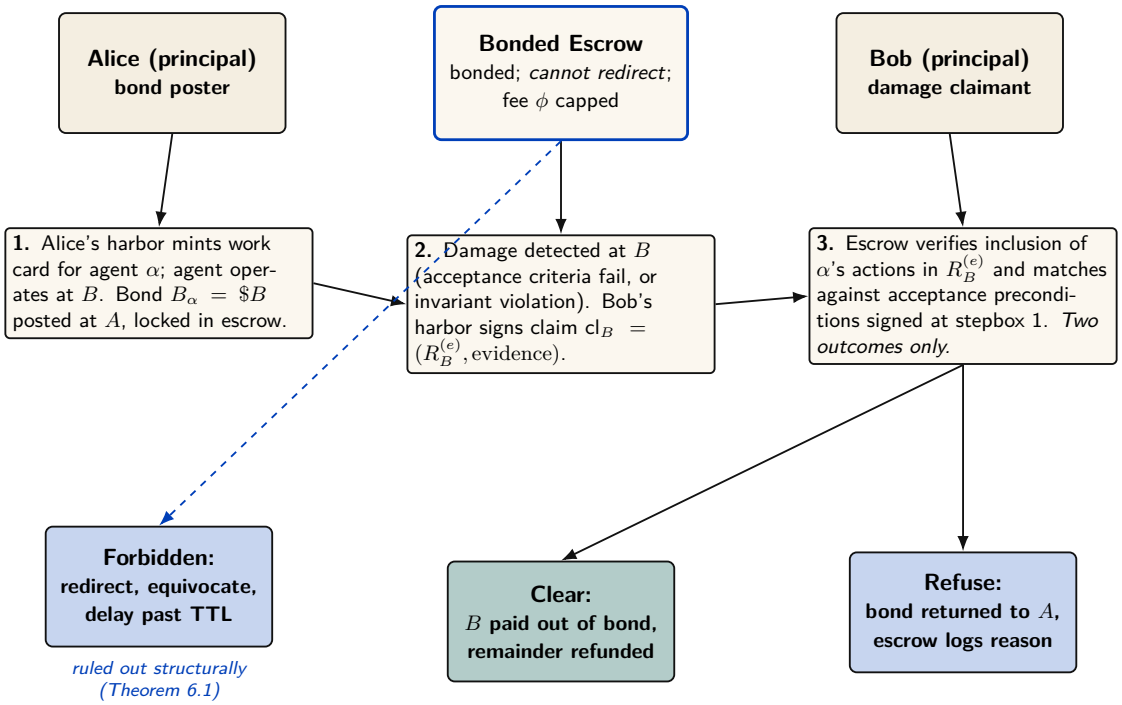


Figure 5: Cross-harbor settlement protocol. The escrow is a third party in the sense that neither $A$ nor $B$ alone can compel an outcome, but it is *not* a trusted third party in the operational sense: the protocol restricts its decision space to two outcomes (clear or refuse) and prices its maximum extractable fee $\phi$ up front. Redirecting funds, equivocating about a clearing decision, or delaying past TTL are structurally forbidden by the property checks in Theorem 6.1. The escrow is a *bounded* third party — the bond it holds plus the cap on $\phi$ make its misbehavior bounded-loss rather than catastrophic.

## 6.1 The bounded escrow

We introduce a third party — the escrow — but we restrict its role structurally. The escrow holds the bond. Its decision space is exactly two outcomes: clear (pay $B$ out of the bond, refund remainder to $A$) or refuse (return the bond to $A$, log the reason). The escrow cannot:

- Redirect funds to anyone other than $A$ or $B$.
- Equivocate about its clearing decision (publish "cleared" to one party and "refused" to another).
- Delay past the TTL of the bond.
- Extract more than a pre-agreed fee $\phi$ which is capped at issuance.

These restrictions are the structural bounds. They are not enforced by trust; they are enforced by the protocol's invariants, which we verify below.

**Theorem 6.1** (Bounded Escrow)**.** Under the cross-harbor settlement protocol of Figure 5, the escrow's worst-case extractable value from any single settlement is bounded by $\phi$ (the pre-agreed fee), regardless of the escrow's strategy.

*Sketch.* The clear and refuse paths both produce a witness-logged settlement record. Redirect, equivocate, and delay are detectable by absence of a settlement record matching the expected pattern within TTL; the protocol invariant is checked by both $A$ and $B$ on their side, and a missing or contradictory record triggers a slashing of the escrow's own bond posted at federation admission (§7). The fee $\phi$ is bound into the escrow's signed acceptance at the start of the settlement; the escrow cannot extract more without violating its own signed commitment, which is also witness-logged. Worst-case extraction is therefore $\phi$. □

The escrow is a *trusted-but-bounded* third party. We are honest that this is not a trustless construction in the HTLC sense [15]. The proposal's open question of whether truly trustless settlement is possible for non-fungible reputation-priced bonds remains open (§12); we conjecture it is not, and we treat the bounded-escrow design as the right pragmatic point in the design space.

### 6.2 Cross-harbor conservation

The Bonded Commons paper proved a single-machine Conservation Theorem: the total bonded value in the system never changes by surprise; bonds are only created (by posting), destroyed (by clearing), or rolled (refunded and re-posted). We extend this to the federation.

**Property 6.1** (Cross-Harbor Conservation)**.** At any moment, the sum of bond values across the federation equals:

$$\sum_{X \in F} \text{posted}_X + \sum_{X \in F} \text{in-escrow}_X - \sum_{X \in F} \text{cleared}_X - \sum_{X \in F} \text{refunded}_X$$

where the sums are taken over all federated harbors $X$, and each term is computed against the witness log up to the current epoch.

This is the federation-layer analogue of Conservation in Bonded. The intra-harbor Conservation invariants hold per-harbor (Bonded gives the proofs); the cross-harbor property adds the constraint that bonds in transit between harbors — i.e., posted at $A$, locked in escrow, awaiting clearance against $B$ — are accounted for as in-escrow rather than as either posted at $A$ or cleared to $B$. The escrow's role as the holder of in-flight bonds is what makes the conservation accounting tractable.

### 6.3 Settlement protocol

Figure 5 shows the three-step protocol. To unpack the figure:

1. **Bond post.** Alice's harbor mints a work card for agent $\alpha$; agent operates at $B$. Bond $B_\alpha = \$B$ is posted at $A$ and locked in the escrow. The bond's amount and TTL are signed into both $A$'s and $B$'s witness-log roots so the federation knows the bond exists.
2. **Damage claim.** Damage detected at $B$ (acceptance criteria fail, or invariant violation). Bob's harbor signs claim $\mathsf{cl}_B = (R_B^{(e)}, \text{evidence})$ where the evidence is a Merkle inclusion proof against $R_B^{(e)}$. The claim is sent to the escrow.
3. **Verification and clearance.** Escrow verifies inclusion of $\alpha$'s actions in $R_B^{(e)}$, matches against the acceptance preconditions that were signed at step 1, and produces one of two outcomes: *clear* (pay $B$ out of bond, refund remainder to $A$) or *refuse* (return bond to $A$, log reason). Both outcomes are recorded in the witness log.

The protocol is *atomic* in the conservation sense: either the bond is forfeited at $A$ and the damaged party at $B$ is paid, or neither happens. Partial clearance is not in the decision space (Theorem 6.1).

### 6.4 Fee structure and economic discipline

The escrow's fee $\phi$ is bounded but non-zero. In the competitive-insurance market sense of Bonded (the Youle contribution in the companion paper), $\phi$ is a market clearing price for the service of holding the bond and verifying the inclusion proof. We do not solve the market here; the Youle contribution from Bonded gives the mechanism, and we conjecture (open) that it extends cleanly to the cross-harbor case when the escrow population is large enough.

We are explicit that for small federations (two or three harbors), $\phi$ is likely set by a posted price rather than auctioned. The mechanism design here is identical in shape to Bonded's, just lifted to the federation layer.

# 7 Federation Admission

The Federated Harbor is not a permissionless market in identities. A new harbor enters the federation through an admission ceremony. The ceremony is invitation-bounded by design: Sybil attacks against the federation [16] are costly for an attacker precisely because each identity requires a real admission event.

## 7.1 The bonded-sponsor pattern

A new harbor $N$ with no prior reputation cannot price its own bond fairly under competitive insurance — the market lacks data on $N$'s risk. The bonded-sponsor pattern resolves this:

1. An existing harbor $S$ (the sponsor) posts a bond on $N$'s behalf, sized to cover the federation's worst-case loss from $N$ misbehaving during a probationary epoch.
2. $N$ joins the federation under probation; its cards are accepted, its evidence trail is witness-logged, but its sponsor bond is at risk.
3. If $N$ misbehaves during probation — forges signatures, equivocates on roots, refuses to honor settled bonds — the sponsor bond is forfeit. The sponsor has full incentive to vet $N$ before sponsoring.
4. At the end of probation, $N$'s own reputation is sufficient to price its own bond, and the sponsor bond is released.

This is the federation-layer analogue of competitive insurance from Bonded (the Youle contribution in the companion paper): instead of an authority picking who joins, the sponsor market discovers it. The cost of being a sponsor is real — forfeit risk during probation — and the benefit is the network effect of having $N$ in the federation. We expect (open) that for federations with many candidate sponsors and clear-eyed risk pricing, the equilibrium is reasonably efficient. We do not prove it.

## 7.2 Cold-start failure modes

A new harbor that cannot find a sponsor cannot join. This is a real limitation. In the worst case, the federation collapses to invitation-only — which is the failure mode of every interesting federated identity system from PGP web-of-trust [18] to SAML [17]. The structural pushback is:

- *Many sponsors.* The market is many-to-many; a new harbor needs only one sponsor. As long as sponsors are paid (via fees or reputation), the supply is nonzero.
- *Small probationary scope.* New harbors during probation are restricted from large-stake operations; the sponsor's risk is bounded.
- *Witness-log permanence.* The admission record is permanent; once a harbor has graduated probation, it does not need a sponsor again.

We are honest that this is engineering discipline, not theorem. The historical evidence on whether invitation-bounded federations stay open is mixed [17]; the Federated Harbor's pushback against centralization is to make the cost of being a centralizing hub bounded above by the cost of running mirrored witness logs, which is small. Whether that pushback is enough is empirical.

# 8 Worked Example

The four primitives are easier to read against a concrete case. We work the example end-to-end: two organizations, three machines, one shared project. The example is intentionally familiar; nothing in it is novel except the way the primitives compose.

### 8.1 Setup

Two organizations: Acme Robotics (Alice's employer) and Beta Vision (Bob's employer). The shared project is a perception-and-control stack: Acme owns the controller, Beta owns the vision system, both run against a shared staging environment that lives on a third machine maintained by neither.

Three machines: Alice's laptop (harbor $A$, controller code, controller test database), Bob's laptop (harbor $B$, vision code, vision test database), staging-server (harbor $S$, shared staging database, no agents resident).

Each harbor has gone through the admission ceremony of §7. Acme's harbor was sponsored by an existing partner harbor at the open-source consortium that hosts the staging server; Beta's was sponsored by Acme after a six-month probationary period during which Acme's risk officer watched Beta's evidence trail closely. The staging server harbor is sponsored by the consortium directly and is treated as a known-honest verifier for the purposes of this example.

### 8.2 The agent task

Acme's agent — call it Controller-7 — needs to run a schema migration on the staging database to add a column for a new sensor type. The migration is destructive (drops and recreates the column); if it goes wrong, the staging database is unusable for the joint demo scheduled three hours later.

Controller-7's local card $C_A$ authorizes `db:migrate` against the controller test database, attenuated through Acme's risk policy ($\text{att}_A$ = "only against tables in the `controller` schema; only between 9am and 5pm; only if a human operator has approved within the last 30 minutes"). The card is valid at $A$, but the database it needs to migrate is at $S$.

### 8.3 Step 1: Cross-harbor transfer

Controller-7 invokes $\mathsf{xfer\_req}(C_A, S)$ on $D_A$. $D_A$ constructs the envelope:

$$\mathsf{xfer\_offer}\langle C_A, \text{att}_A, R_A^{(e)}\rangle \ \text{ signed under } \ sk_A$$

and sends it to $D_S$ (the staging-server harbor). $D_S$ verifies the signature against $A$'s public key as recorded in the witness log, verifies that $R_A^{(e)}$ is the current epoch root for $A$, and applies its own attenuation $\text{att}_S$ ("only the `controller` schema; only after a backup snapshot has been taken; only if no other migration is in flight"). The composed attenuation produces $C'_S = \text{att}_S(\text{att}_A(C_A))$, a card valid only at $S$, signed under $sk_S$.

$D_S$ delivers $C'_S$ to Controller-7. Controller-7 takes the snapshot, then runs the migration against $S$. The migration's actions are recorded in $S$'s Merkle forest and roll up into $R_S^{(e)}$.

### 8.4 Step 2: Damage detection

The migration runs cleanly. Two hours later — well after the agent has moved on — Beta's vision system runs an integration test against the staging database and observes that the new column has the wrong type (Acme assumed `float32`; Beta needs `float64` for the sensor data they will produce). The integration test fails.

$D_B$ (Beta's harbor) verifies the failure against its local invariants and produces a damage claim:

$$\mathsf{cl}_B = (R_B^{(e+1)}, \text{evidence})$$

where the evidence is a Merkle proof that Beta's integration tests against the staging database failed because of a schema mismatch. The claim is sent to the escrow.

### 8.5 Step 3: Settlement

Acme posted a bond when Controller-7 was authorized to run the migration. The bond is held in the federation escrow with a TTL of 24 hours — long enough for downstream consequences to surface, short enough not to lock collateral indefinitely.

The escrow verifies $\mathsf{cl}_B$: it checks the inclusion proof against $R_B^{(e+1)}$ in the witness log, matches the claim against the acceptance preconditions signed at bond-post time (which named the staging schema as a covered surface), and decides: *clear*. Acme's bond is debited; Beta receives the cleanup-cost portion; the remainder is refunded to Acme. The settlement record is written to the witness log.

### 8.6 Step 4: Revocation propagation

At this point, Acme's risk officer revokes Controller-7's `db:migrate` card — not as punishment, but because the failure indicates Controller-7's schema-design heuristic was wrong and the card should not be reused until that is fixed. The revocation is added to $A$'s local cuckoo filter and $A$'s next epoch root $R_A^{(e+2)}$ is published to the witness log. By Theorem 5.1, within $\Delta(1 + \ln m)$ time units, every federated harbor's filter contains the revocation. By that point, every $C'_S$ derived from Controller-7's card is invalid at $S$ as well.

### 8.7 What this example shows

The example is small and the failure is real. The point of working it is to show that each of the four primitives — transfer, revocation, settlement, admission — is doing its job at exactly one step, and that the composition is straightforward when each primitive is honest about what it covers.

Notice in particular what *did not* happen:

- Neither $D_A$ nor $D_B$ trusted the other's signing key as a root. The transfer worked through the witness-logged public keys, not through a chain of trust.
- The bond was not held by either harbor; it was held by an escrow whose role was structurally bounded.
- The settlement was not negotiated peer-to-peer; it followed a fixed protocol whose two outcomes were named in advance.
- The revocation propagated through gossip with a named convergence bound; no consensus protocol was invoked.

Each of these is a deliberate design choice. The Federated Harbor's architecture is not heroically more powerful than alternatives; it is heroically more honest about what is happening at each step.

## 9 Formal Model

The formal substrate of this paper is two-layer: ProVerif models for the cryptographic protocols (cross-realm authenticity, attenuation, settlement no-redirect) and TLA+ extensions of Bonded's Conservation specification for the cross-harbor accounting invariants.

### 9.1 ProVerif extensions to Anchor

The Anchor Protocol's three-phase models are extended with a second signing key and an epoch-root binding. The cross-realm authenticity query becomes:

```
query b: id, ha: id, hb: id, cap: capability, e: epoch;
  event(AcceptedAtB(b, hb, cap, e))
  ==> (event(IssuedAtA(b, ha, cap)) &&
```

```
        event(EnvelopeAtoB(ha, hb, cap, e)) &&
        event(RootCurrent(ha, e))).
```

Listing 1: Cross-realm authenticity query (sketch)

This says: if $B$ accepts a cross-realm card for capability `cap` at epoch $e$, then there exists a corresponding issuance event at $A$, an envelope event from $A$ to $B$ binding the same capability and epoch, and an event confirming that $R_A^{(e)}$ is the current root. The query holds against the Dolev–Yao attacker model under the standard assumption of Ed25519 EUF-CMA security.

We are honest that the model is drafted and the proof script is in progress. The full mechanization is named open work; see Appendix A for the status table.

## 9.2 TLA+ extension for cross-harbor conservation

The Bonded Commons paper's Conservation specification models a single-machine bond pool. The Federated Harbor extension adds a per-harbor pool, an escrow pool, and an inter-harbor transit relation:

```
CONSTANTS Harbors, MaxBonds, Capabilities
VARIABLES PostedAt, InEscrow, ClearedTo, RefundedTo, WitnessLog

TypeOK ==
  /\ PostedAt   \in [Harbors -> Nat]
  /\ InEscrow   \in [Harbors -> Nat]
  /\ ClearedTo  \in [Harbors -> Nat]
  /\ RefundedTo \in [Harbors -> Nat]
  /\ WitnessLog \in Seq([harbor: Harbors, epoch: Nat, root: STRING])

Conservation ==
  \A h \in Harbors:
    PostedAt[h] + InEscrow[h] - ClearedTo[h] - RefundedTo[h] >= 0

Spec == Init /\ [][Next]_<<PostedAt, InEscrow, ClearedTo,
                           RefundedTo, WitnessLog>>
```

Listing 2: Cross-Harbor Conservation specification (sketch)

The full specification (which we omit for space; see Appendix C) extends Bonded's *Next* relation with three actions: `PostBond`, `TransferToEscrow`, and `Settle`, the last of which has two outcomes (`Clear` and `Refuse`) matching the bounded-escrow theorem. The Apalache symbolic model checker handles the resulting state space at $|F| = 4$ harbors with bond depth 16; we run it at this scale to establish that the Conservation invariant is preserved across the federation. Beyond that scale, we lean on the structural argument rather than full model checking.

## 9.3 What is mechanized and what is structural

We are explicit about which claims live in which band (Figure 2).

**Mechanized (sage).** Single-harbor capability authenticity (inherited from Anchor); cross-realm authenticity under the witness-log freshness assumption (draft, **partial**); attenuation monotonicity across the boundary (Lemma 4.1, **partial**under same caveat); cross-harbor Conservation invariant (TLA+, model-checked at $|F| \leq 4$).

**Bounded (amber).** Federated revocation convergence in expected $\Delta(1 + \ln m)$ time (Theorem 5.1); cross-witness equivocation detection in $O(\log m)$ rounds; bounded-escrow extractable value at most $\phi$ (Theorem 6.1, structural argument).

**Open (cobalt).** Multi-principal correlated-equilibrium extension of Bonded; trustless settlement for non-fungible reputation-priced bonds (conjectured impossible); folk-theorem cartel-resistance bound at the federation layer.

# 10 Failure Modes

A federation paper that does not engage honestly with the historical failure modes of federated systems is unserious. The literature is heavy with examples: SAML federations that concentrated on a small set of identity providers [17]; PGP web-of-trust that never achieved meaningful adoption outside enthusiast circles [19]; Sigstore's gravitation toward a single deployment of Fulcio [10]. We engage with three failure modes specifically.

## 10.1 Centralization gravity

Federated identity systems have a well-documented tendency to centralize on a small number of trust anchors within five to ten years of deployment [17]. The Federated Harbor's structural pushback is the replicable witness log: any harbor can run its own witness instance, any harbor can audit any other instance's roots, and equivocation between witnesses is detectable. This makes it more expensive to be a centralizing hub than to be a peer.

The pushback is not theorem; it is design discipline. We are honest about this in Figure 2 (cobalt band). The historical evidence is mixed. Tailscale [25] has remained operationally centralized on its own coordination server despite Headscale [26] being a viable alternative; in-toto [11] has stayed decentralized but is consumed mostly by SLSA [12] which has Google as its de facto center. Time will tell which pattern the Federated Harbor exhibits.

## 10.2 Equivocation by a malicious witness

A malicious witness log instance could serve different views to different auditors. The cross-witness auditor pattern of §2.3 detects this in expected $O(\log m)$ rounds; the witness's bond is slashed on confirmation. The risk is the window between equivocation and detection.

For high-stakes settlements, harbors can require multiple witness signatures before treating a root as authoritative, in the Certificate Transparency two-SCT pattern [9]. This trades latency for assurance; the design space is well-explored in the CT literature and we adopt the same techniques.

## 10.3 Partition tolerance

The federation does not run consensus; partition tolerance is per-protocol. Capability transfer cannot proceed during a partition between $A$ and $B$ (no surprise; cross-realm authentication requires both daemons to reach the witness log). Revocation gossip continues within each partition; convergence is bounded by the partition's local size, not the global federation size. Settlement is parked at the escrow until the partition heals.

The structural posture is that partitions are operationally common and the federation should degrade gracefully rather than fail fast. We inherit this discipline from Anchor's intra-mesh design [1]; the federation extension is straightforward but adds no novel mathematics.

# 11 Related Work

The Federated Harbor sits at the intersection of four bodies of work. We are concrete about where we draw on each and where we depart.

**Federated identity.** The OpenID Federation 1.0 specification [21] is the closest existing analogue: each entity publishes a signed Entity Statement, trust chains assemble by walking up to a Trust Anchor, trust marks express property claims. The Federated Harbor is structurally similar in the trust-anchor pattern but differs in two places: (a) we add a capability/attenuation layer (OpenID Federation is identity-only), and (b) we do not assume HTTP-based fetching of entity statements, since Port Daddy daemons cannot reliably expose HTTP endpoints (NAT, sleep, etc.). We draw also on SAML/Shibboleth historical experience [17] as cautionary lineage; the centralization gravity we discuss in §10 is exactly the SAML-era failure mode.

**Capability-based federation.** Macaroons [4] are the foundational capability-with-attenuation construction; our transfer ceremony extends Macaroon-style third-party-caveat chains across an administrative boundary. UCAN [20] is the current production-deployed answer to "Macaroons with public-key principals," and we treat it as the operationally most relevant existing system; the Federated Harbor differs in keeping the daemon as a structural commons authority rather than going fully peer-to-peer, and in pricing delegation through insurance-priced bonds rather than pure capability constraints. The object-capability lineage [6, 5] continues to be relevant: cross-realm operations are exactly a mutual-suspicion scenario in Miller's vocabulary, and his defensive-consistency framework is the right lens.

**Transparency logs.** Certificate Transparency [9] is the model; the federated witness log of §2.3 is CT applied to capability tokens. Sigstore [10] and Rekor are the production instantiation we rhyme with most heavily; in-toto [11] and SLSA [12] are the supply-chain analogues that compose with the Federated Harbor for the "poisoned plugin under a bonded principal" case but do not close it. The Crosby–Wallach history-tree [13] is the formal substrate of every modern transparency log including ours.

**Cross-domain settlement.** Herlihy's atomic cross-chain swap [15] is the protocol template; the cross-harbor settlement protocol of §6 mirrors the structure but specializes for non-fungible reputation-priced bonds. We are explicit that our settlement is not trustless in Herlihy's sense; it admits a trusted-but-bounded escrow. Whether trustless is achievable in our setting is open (§12). The IBC protocol [22] is the most production-deployed cross-domain communication protocol that is not on Ethereum, and its connection/channel abstraction influenced our handshake; we differ in not requiring consensus underneath.

# 12 Limitations and Open Questions

We close with the honest open frontier. These are not throwaway caveats; they are the questions the paper does not answer, named precisely enough to motivate subsequent work.

## 12.1 The multi-principal correlated-equilibrium extension

The Bonded Commons paper argued that single-machine coordination is a correlated equilibrium with the daemon as the correlation device [2]. Across two harbors with two daemons, the correlation device fragments. Two candidate resolutions remain unresolved:

- The federated witness log *is* the correlation device, and individual harbors are local enactors. This would extend Bonded's result intact.
- The cross-harbor case is a partition of the correlated-equilibrium concept into intra-harbor (correlated) and inter-harbor (Nash with shared signal). This would weaken Bonded's result at the federation layer.

We do not know which is right. The proposal that scoped this paper [3] named Thomas Youle as the load-bearing collaborator for this question (his contribution to Bonded is the closest existing work). The question stays open here.

### 12.2 Trustless settlement for non-fungible bonds

HTLC-style atomic swaps [15] achieve trustless settlement for fungible assets with a shared hash preimage. The Federated Harbor's bonds are non-fungible (reputation-priced under competitive insurance) and may need a trusted-but-bounded escrow rather than a trustless one. We adopt the bounded-escrow design (§6) but conjecture — without proof — that fully trustless settlement is impossible for this asset class. A clean negative result would be a worthwhile separate paper.

### 12.3 Cartel formation across federations

Bonded's folk-theorem cartel-resistance argument applies inside a single machine. Between machines, cartel formation is structurally easier: colluders can coordinate offline and present a unified front to the witness log, which records that all harbors agreed — a behavior consistent with both honest agreement and collusion. We owe either a strengthening of the bonded mechanism that resists cross-harbor cartels, or a precise statement of the weakening. We have neither.

### 12.4 Cold-start admission and the invitation-only failure mode

The bonded-sponsor pattern of §7 requires a new harbor to find a sponsor willing to underwrite its probationary risk. In practice, this market may be thin — the historical evidence on whether invitation-bounded federations stay open is mixed [17]. We treat the structural pushback (many sponsors, small probationary scope, witness-log permanence) as engineering discipline, not as theorem.

### 12.5 Equivocation propagation speed

Theorem 5.1 gives expected $O(\log m)$ detection of cross-witness equivocation. The constant is empirical; for very small federations ($m \leq 5$) the bound is loose, and for very large ones ($m \geq 1000$) we have no operational data. The window between equivocation and detection is also the attacker's window; a tight quantitative bound would tighten the pricing of the witness bond at admission. We name this as open optimization rather than open feasibility.

### 12.6 What this means for the reader

A federation paper with five named open questions is doing its job. The point of the paper is to draw the new boundary cleanly so the open problems are visible; if every claim in this paper were closed, there would be no third paper to write. The honest read is that the Federated Harbor is closer to a research agenda than to a deployed product, and the prior two papers (Anchor, Bonded) have a tighter ratio of theorem to conjecture for the same reason. The substrate is real; the federation layer is in progress.

> *The point of the paper is to draw the new boundary cleanly so that the unsolved problems are visible rather than hidden.*

## 13 Conclusion

The two-machine problem is the structural sequel to the local-swarm problem of Anchor and the trust-as-infrastructure argument of Bonded. The proposed answer is not a new substrate but a clean extension: keep each daemon sovereign inside its own machine, add a shared witness log that no harbor controls, hang revocation gossip and a bounded settlement escrow off the witness log, and gate federation admission on bonded sponsorship rather than permissionless entry. Each primitive does exactly one job. Each primitive is honest about what it does not do.

The proof posture matches the prior papers' discipline. Where we mechanize, we mechanize (with the same ProVerif and TLA+ infrastructure). Where we bound a threat, we name the bound and price the bond against it. Where we leave a question open, we name it. The threat-band diagram of Figure 2 is the most concentrated summary of where each claim lives.

What this paper buys is a clean federation surface that composes with the substrate already in production. Two harbors that have never met can run a shared project. Bonds posted at one harbor clear against damage measured at another. Revocations issued at one harbor reach every harbor in the federation within a named convergence bound. New harbors join under bonded sponsorship without forcing the federation into either an invitation-only failure mode or a Sybil-prone permissionless one. None of these is heroic in isolation. The contribution is the composition.

The remaining open work — the multi-principal correlated-equilibrium extension, the trustless-settlement impossibility conjecture, the cartel-resistance bound at the federation layer — is genuine open work. It is not minor. The paper is honest about that. The federation is real; the theory of the federation is in progress.

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# A Verification Status

**Artifact registry.** The cross-paper status table in Bonded Commons [2] carries the items that are shared across the series; this appendix lists only the items specific to the Federated Harbor.

Table 1: Federated Harbor verification status. **closed**: model verified and runtime conformance test passes. **partial**: design specified, mechanization in draft. **open**: known unmechanized; threat band visible in Figure 2 (cobalt).

| Claim | Method | Result | Artifact |
|---|---|---|---|
| Single-harbor capability authenticity (inherited) | ProVerif 2.05 | **closed** | Anchor `harbor_card_v{1,2,3}*.pv` |
| Cross-realm authenticity (§4) | ProVerif (draft) | **partial** pending witness-log freshness assumption | `fh-xfer.pv` (draft) |
| Attenuation monotonicity across boundary (Lem. 4.1) | ProVerif + Anchor sub-result | **partial** | `fh-att.pv` (draft) |
| Federated revocation convergence (Thm. 5.1) | Analytic, building on [14] | **partial** | `fh-conv.tex` (proof sketch) |
| Cross-witness equivocation detection in $O(\log m)$ | Analytic, CT-style argument | **partial** | — |
| Bounded-escrow extractable value (Thm. 6.1) | Protocol invariants + runtime check | **partial** | `fh-escrow.pv` (draft) |
| Cross-harbor Conservation (§9.2) | TLA+ / Apalache, $\lvert F\rvert \leq 4$ | **partial** model-checked at scale | `CrossHarborConservation.tla` |
| Trustless settlement for non-fungible bonds | — | **open** conjectured impossible | — |
| Multi-principal correlated-equilibrium extension | — | **open** | — |
| Cross-federation cartel-resistance bound | — | **open** | — |

**What this table means.** Every row in the **partial** band is committed to and on the active workplan. Every row in the **open** band is a research question we cannot resolve from the desk; the proposal [3] names collaborators whose contributions would be load-bearing. We err toward listing fewer items as closed rather than more, in the same posture as Anchor and Bonded.

# B ProVerif Models (draft)

For brevity we omit the full model text in this pre-print; the draft `fh-xfer.pv` extends Anchor's `harbor_card_v3_delegation.pv` with a second signing key $sk_B$, an epoch root binding, and a cross-realm authenticity query of the shape sketched in §9.1. The full model will accompany the revised version once the witness-log freshness assumption is mechanized rather than assumed.

# C TLA+ Specification (draft)

The full `CrossHarborConservation.tla` extends Bonded's Conservation specification with per-harbor pools, an escrow pool, and an inter-harbor transit relation. Apalache symbolic model-checking at $\lvert F\rvert = 4$ harbors and bond depth 16 establishes that Conservation holds across all

reachable states under the transition relation. The full specification accompanies the revised version.